A:Transformation is all you need (15)

1. Eigenvalues and eigenvectors. $(3 + 3 = 6)$
   a. Without computing the eigenvalues and vectors of a 3D translation matrix, use the definition of eigenvalues and vectors and geometric intuition to describe how they would/should look like **(1.5+1.5)**
   b. Compute the eigenvalues and eigenvectors of a 3D translation matrix. Give the full characterization **(1.5+1.5)**

2. Properties of transformations $(3 + 3 + 3 = 9)$
   a. Do all orthonormal matrices represent strictly rotations? What about orthogonal matrices?
   b. Prove that perspective projection does not preserve parallel lines.
   c. If a 2D translation matrix represented in homogenous coordinates is treated as a 3D transformation matrix, what transformation does it correspond to? Substantiate.

B:Flatland (7)

1. Design a mid-point criterion based algorithm to draw a parabola given by the equation $y^2 = 4ax$. Derive the decision parameter, base case and the incremental updates, finally write the algorithm. **(5)**

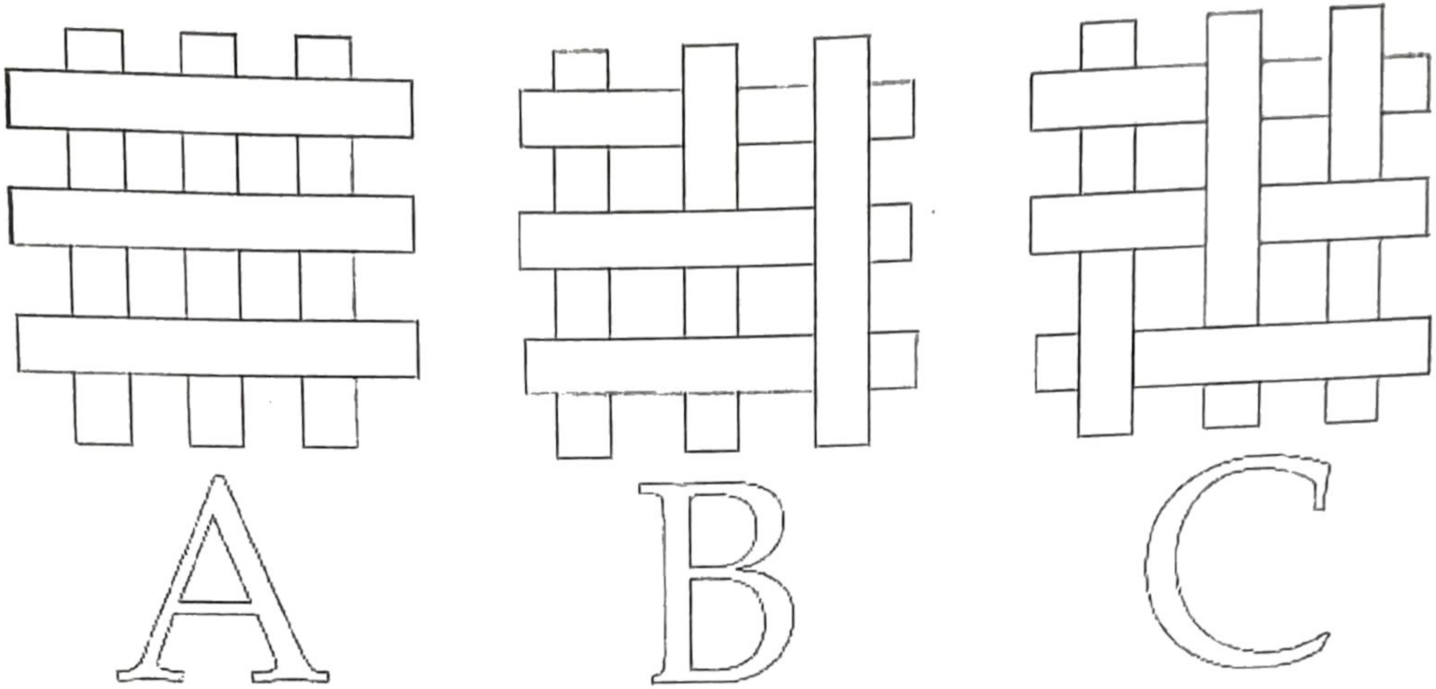
2. Clipping **(2)**
   a. Draw convex quadrilaterals so that Sutherland-Hodgeman clipping algorithm with a rectangular clipping window results in polygons having 5, 8 sides.

C: Let there be light! (13)

1. Compute the surface normal at a point (x, y, z) on an ellipsoid given by $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$? (Hint: It is related to gradient) **(2)**

2. In the Blinn-Phong illumination model. Show that if v lies in the same plane as l, n, and r, then the halfway angle satisfies $2\psi = \varphi$. (ψ is the angle between h and n, φ is the angle between v and r) Provide the motivation for the half angle. **(2+2 = 4)**

3. Which algorithm will you use to draw the primitives in the image below named A,B,C? Substantiate.**(3)**



4. Solve the discrepancies
   a. Assume you have two instances of the same triangle drawn side by side (same material properties, one is translated wrt other, no rotation/scaling is applied) . You see that one has a bright color while the other is completely dark. Describe the reason for this **(1)**
   b. A model with no holes is constructed with triangles, but you see a lot of holes when you render it. What might be wrong? **(1)** how can you fix this? **(2)**